

1. Launch one aws ec2 and install jenkins.
1. Region : [N. Virginia](#) us-east-1
2. Instance Name : EC2_4Jenkins
3. Key Pair :my_server.pem
4. Security Key : launch-wizard-1

AWS EC2 – Launch Instance and Install Jenkins

1. Launch an EC2 Instance

1. Login to the **AWS Management Console**.
2. Navigate to **EC2** → **Instances**.
3. Click **Launch Instance**.
4. Enter the instance name, for example:

Jenkins-Server

5. Select an Ubuntu AML.
6. Select an instance type, for example:

t2.micro

(Use the instance type available/appropriate for your AWS account.)

7. Create or select an existing **Key Pair**.
8. Configure the Security Group with the required inbound rules:
 - **SSH – TCP 22** → My IP
 - **Custom TCP – 8080** → My IP / required source
9. Keep the required storage configuration.
10. Click **Launch Instance**.

2. Connect to the EC2 Instance

1. Select the newly created EC2 instance.

2. Click **Connect**.
3. Select **SSH Client**.
4. Connect using the provided SSH command.

Example:

```
ssh -i "jenkins-key.pem" ubuntu@<PUBLIC-IP>
```

After successful login:

```
ubuntu@ip-172-31-21-0:~$
```

3. Update Ubuntu Packages

Run:

```
sudo apt update
```

Then:

```
sudo apt upgrade -y
```

4. Install Java

Jenkins requires Java to run.

Install Java 21:

```
sudo apt install fontconfig openjdk-21-jre -y
```

Verify Java:

```
java -version
```

Example:

```
openjdk version "21..."
```

5. Add Jenkins Repository Key

Remove any old Jenkins repository configuration if present:

```
sudo rm -f /etc/apt/sources.list.d/jenkins.list  
sudo rm -f /usr/share/keyrings/jenkins-keyring.asc
```

Create the keyrings directory:

```
sudo mkdir -p /etc/apt/keyrings
```

Download the Jenkins repository signing key:

```
sudo wget -O /etc/apt/keyrings/jenkins-keyring.asc \  
https://pkg.jenkins.io/debian-stable/jenkins.io-2026.key
```

6. Add Jenkins Repository

Run:

```
echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc] \  
https://pkg.jenkins.io/debian-stable binary/ | \  
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
```

7. Update Package Repository

```
sudo apt update
```

This downloads the latest package information, including Jenkins.

8. Install Jenkins

```
sudo apt install jenkins -y
```

9. Start Jenkins Service

Start Jenkins:

```
sudo systemctl start jenkins
```

Enable Jenkins to start automatically after reboot:

```
sudo systemctl enable jenkins
```

Or both can be done together:

```
sudo systemctl enable --now jenkins
```

10. Check Jenkins Status

Run:

```
sudo systemctl status jenkins
```

Expected:

```
Active: active (running)
```

Press **q** to exit the status screen.

11. Get Jenkins Initial Admin Password

Run:

```
sudo cat /var/lib/jenkins/secrets/initialAdminPassword
```

Copy the displayed password.

12. Access Jenkins from Browser

Open:

```
http://<EC2-PUBLIC-IP>:8080
```

Example:

```
http://54.xx.xx.xx:8080
```

The **Unlock Jenkins** page will appear.

Paste the initial administrator password.

13. Complete Jenkins Setup

1. Click **Install suggested plugins**.
 2. Create the Jenkins administrator user.
 3. Configure the Jenkins URL if required.
 4. Click **Save and Finish**.
 5. Jenkins dashboard will open.
-

Final Verification

Check the Jenkins service:

```
sudo systemctl status jenkins
```

Check Java:

```
java -version
```

Check Jenkins port:

```
sudo ss -lntp | grep 8080
```

Expected Jenkins port:

```
8080
```

5. Screenshots for reference

aws

Search

[Alt+S]

United States (N. Virginia)

chandrashekar (986881030460)

chandrashekar

EC2 > Instances > Launch an instance

It seems like you may be new to launching instances in EC2. Take a walkthrough to learn about EC2, how to launch instances and about best practices

Take a walkthrough Do not show me this message again

Launch an instance

Amazon EC2 allows you to create virtual machines, or instances, that run on the AWS Cloud. Quickly get started by following the simple steps below.

Name and tags

Name

EC2-4jenkins

Add additional tags

Application and OS Images (Amazon Machine Image)

An AMI contains the operating system, application server, and applications for your instance. If you don't see a suitable AMI below, use the search field or choose [Browse more AMIs](#).

Search our full catalog including 1000s of application and OS images

Recents

Quick Start

Amazon Linux

aws

macOS

Mac

Ubuntu

Canonical Ubuntu

Windows

Microsoft

Red Hat

Red Hat

SUSE Linux

SUSE

Debian

debian

Browse more AMIs

Including AMIs from AWS, Marketplace and the Community

Amazon Machine Image (AMI)

Ubuntu Server 26.04 LTS (HVM), SSD Volume Type

ami-0b6d9d3d33ba97d99 (64-bit x86_64) / ami-0bc7f2dbdccb55303 (64-bit Arm64)

Free tier eligible

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-0b6d9d3d33ba97d99

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

aws

Search

[Alt+S] Ask Amazon Q

United States (N. Virginia)

chandrashekar (986881030460)

chandrashekar

EC2 > Instances > Launch an instance

Instance type

Info | Get advice

Instance type

t3.micro

Family: t3 2 vCPU 1 GiB Memory Current generation: true On-Demand Linux base pricing: 0.0104 USD per Hour On-Demand Windows base pricing: 0.0196 USD per Hour On-Demand Ubuntu Pro base pricing: 0.0139 USD per Hour On-Demand SUSE base pricing: 0.0104 USD per Hour On-Demand RHEL base pricing: 0.0392 USD per Hour

Free tier eligible

All generations

Compare instance types

Additional costs apply for AMIs with pre-installed software

Key pair (login)

Info

You can use a key pair to securely connect to your instance. Ensure that you have access to the selected key pair before you launch the instance.

Key pair name - required

my_server

Create new key pair

Network settings

Info

Network

vpc-09ace864d17ee9449

Subnet

No preference (Default subnet in any availability zone)

Auto-assign public IP

Enable

Firewall (security groups)

Edit

Summary

Number of instances

1

Software Image (AMI)

Canonical, Ubuntu, 26.04, amd64...[read more](#)

ami-0b6d9d3d33ba97d99

Virtual server type (instance type)

t3.micro

Firewall (security group)

New security group

Storage (volumes)

1 volume(s) - 8 GiB

Cancel

Launch instance

Preview code

AWS Management Console - EC2 Instances

Instances (1/1) Info

Filter Instance by attribute or tag (case-sensitive)

Name	Instance ID	Instance state	Instance type	Status check	Availability Zone	Public IPv4 DNS	Public IPv4 ...	Elastic
Ec2-4jenkins	i-0f60d65980d510651	Running	t3.micro	Initializing	us-east-1c	ec2-54-242-40-98.com...	54.242.40.98	-

i-0f60d65980d510651 (Ec2-4jenkins)

Details | Status and alarms | Monitoring | Security | Networking | Storage | Tags

Instance summary Info

Instance ID i-0f60d65980d510651	Public IPv4 address 54.242.40.98 open address	Private IPv4 addresses 172.31.21.0
IPv6 address -	Instance state Running	Public DNS ec2-54-242-40-98.compute-1.amazonaws.com open address
Hostname type IP name: ip-172-31-21-0.ec2.internal	Private IP DNS name (IPv4 only) ip-172-31-21-0.ec2.internal	Elastic IP addresses -
Answer private resource DNS name IPv4 (A)	Instance type t3.micro	AWS Compute Optimizer finding -
Auto-assigned IP address	VPC ID	

```
ubuntu@ip-172-31-21-0: ~
HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)
$ chmod 400 "my_server.pem"
HP@DESKTOP-1GQ065F MINGW64 ~/Downloads (master)
$ ssh -i "my_server.pem" ubuntu@ec2-54-242-40-98.compute-1.amazonaws.com
The authenticity of host 'ec2-54-242-40-98.compute-1.amazonaws.com (54.242.40.98)' can't be established.
ED25519 key fingerprint is SHA256:KDW0TEX6fhns0m3cZbArnOewQHUX538XnzUlwMjs7rc.
This key is not known by any other names
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes
Warning: Permanently added 'ec2-54-242-40-98.compute-1.amazonaws.com' (ED25519) to the list of known hosts.
Welcome to Ubuntu 26.04 LTS (GNU/Linux 7.0.0-1006-aws x86_64)

 * Documentation:  https://docs.ubuntu.com
 * Management:    https://landscape.canonical.com
 * Support:       https://ubuntu.com/pro

System information as of Thu Sep 17 07:06:55 UTC 2026

System load: 0.0           Temperature: -273.1 C
Usage of /: 30.5% of 6.61GB Processes: 120
Memory usage: 26%         Users logged in: 0
Swap usage: 0%            IPv4 address for ens5: 172.31.21.0

Expanded Security Maintenance for Applications is not enabled.

0 updates can be applied immediately.

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

The list of available updates is more than a week old.
To check for new updates run: sudo apt update

The programs included with the Ubuntu system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by
applicable law.

ubuntu@ip-172-31-21-0:~$
```

Install Jenkins on EC2

Task: Launch an AWS EC2 instance and install Jenkins.

Steps:

1. Launch Ubuntu EC2 instance.
2. Configure Security Group for SSH (22) and Jenkins (8080).
3. Connect to EC2 using SSH.
4. Update Ubuntu packages.
5. Install Java 21.
6. Add Jenkins repository and signing key.
7. Install Jenkins.
8. Start and enable Jenkins service.
9. Verify Jenkins service status.
10. Retrieve the initial Jenkins administrator password.
11. Access Jenkins using EC2 Public IP on port 8080.
12. Complete the Jenkins initial setup.

```
Setting up openjdk-21-jre:amd64 (21.0.12+8-1~26.04) ...
Processing triggers for libc-bin (2.43-2ubuntu2) ...
Processing triggers for libgdk-pixbuf-2.0-0:amd64 (2.44.5+dfsg-4ubuntu1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-21-0:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-21-0:~$ |
```

```

ubuntu@ip-172-31-21-0:~$ sudo wget -O /usr/share/keyrings/jenkins-keyring.asc \
https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
2026-09-17 07:13:24 -- https://pkg.jenkins.io/debian-stable/jenkins.io-2023.key
Resolving pkg.jenkins.io (pkg.jenkins.io)... 140.248.150.133, 2a04:4e42:79::645
Connecting to pkg.jenkins.io (pkg.jenkins.io)[140.248.150.133]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 3175 (3.1K) [application/octet-stream]
Saving to: '/usr/share/keyrings/jenkins-keyring.asc'

/usr/share/keyrings/jenkins-keyring.asc 100%[=====] 3.10K --.-KB/s in 0s

2026-09-17 07:13:24 (39.6 MB/s) - '/usr/share/keyrings/jenkins-keyring.asc' saved [3175/3175]

ubuntu@ip-172-31-21-0:~$ echo deb [signed-by=/usr/share/keyrings/jenkins-keyring.asc] \
https://pkg.jenkins.io/debian-stable binary/ | sudo tee \
/etc/apt/sources.list.d/jenkins.list > /dev/null
ubuntu@ip-172-31-21-0:~$ sudo apt update
sudo apt install jenkins -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Ign:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:5 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Ign:6 https://pkg.jenkins.io/debian-stable binary/ Release.gpg
Hit:7 https://security.ubuntu.com/ubuntu resolute-security InRelease
Warning: OpenPGP signature verification failed: https://pkg.jenkins.io/debian-stable binary/ Release: The following signatures couldn't be verified because the public key is not available: NO_PUBKEY 7198F4B714BFC68
Error: The repository 'https://pkg.jenkins.io/debian-stable binary/ Release' is not signed.
Notice: Updating from such a repository can't be done securely, and is therefore disabled by default.
Notice: See apt-secure(8) manpage for repository creation and user configuration details.
Package jenkins is not available, but is referred to by another package.
This may mean that the package is missing, has been obsoleted, or
is only available from another source
Error: Package 'jenkins' has no installation candidate
ubuntu@ip-172-31-21-0:~$

```

```

/etc/apt/keyrings/jenkins-keyring.asc 100%[=====] 1.64K --.-KB/s in 0s

2026-09-17 07:24:26 (25.7 MB/s) - '/etc/apt/keyrings/jenkins-keyring.asc' saved [1680/1680]

ubuntu@ip-172-31-21-0:~$ echo "deb [signed-by=/etc/apt/keyrings/jenkins-keyring.asc]" \
https://pkg.jenkins.io/debian-stable binary/ | \
sudo tee /etc/apt/sources.list.d/jenkins.list > /dev/null
ubuntu@ip-172-31-21-0:~$ sudo apt update
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Hit:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:7 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:8 https://pkg.jenkins.io/debian-stable binary/ Packages [31.7 kB]
Fetched 34.6 kB in 0s (139 kB/s)
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-21-0:~$ sudo apt install fontconfig openjdk-21-jre -y
fontconfig is already the newest version (2.17.1-ubuntu1).
openjdk-21-jre is already the newest version (21.0.12+8-1-26.04).
Summary:
Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 173
ubuntu@ip-172-31-21-0:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-21-0:~$ sudo apt install jenkins -y
Installing:
jenkins
Installing dependencies:
net-tools
Summary:
Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 173
Download size: 101 MB
Space needed: 102 MB / 4189 MB available
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64v3 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.568.3 [101 MB]
Fetched 101 MB in 2s (45.8 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 89573 files and directories currently installed.)
Preparing to unpack .../net-tools-2.10-2ubuntu1.amd64v3.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins-2.568.3.all.deb ...
Unpacking jenkins (2.568.3) ...
Setting up net-tools (2.10-2ubuntu1) ...
Setting up jenkins (2.568.3) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' -> '/usr/lib/systemd/system/jenkins.service'.

```

```

Hit:4 http://security.ubuntu.com/ubuntu resolute-security InRelease
Ign:5 https://pkg.jenkins.io/debian-stable binary/ InRelease
Get:6 https://pkg.jenkins.io/debian-stable binary/ Release [2044 B]
Get:7 https://pkg.jenkins.io/debian-stable binary/ Release.gpg [833 B]
Get:8 https://pkg.jenkins.io/debian-stable binary/ Packages [31.7 kB]
Fetched 34.6 kB in 0s (139 kB/s)
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-21-0:~$ sudo apt install fontconfig openjdk-21-jre -y
fontconfig is already the newest version (2.17.1-3ubuntu1).
openjdk-21-jre is already the newest version (21.0.12+8-1-26.04).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 173
ubuntu@ip-172-31-21-0:~$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12+8-1-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12+8-1-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-21-0:~$ sudo apt install jenkins -y
Installing:
  jenkins
Installing dependencies:
  net-tools
Summary:
  Upgrading: 0, Installing: 2, Removing: 0, Not Upgrading: 173
  Download size: 101 MB
  Space needed: 102 MB / 4189 MB available
Get:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute/main amd64 net-tools amd64 2.10-2ubuntu1 [152 kB]
Get:2 https://pkg.jenkins.io/debian-stable binary/ jenkins 2.568.3 [101 MB]
Fetched 101 MB in 2s (45.8 MB/s)
Selecting previously unselected package net-tools.
(Reading database ... 89573 files and directories currently installed.)
Preparing to unpack .../net-tools_2.10-2ubuntu1_amd64.deb ...
Unpacking net-tools (2.10-2ubuntu1) ...
Selecting previously unselected package jenkins.
Preparing to unpack .../jenkins_2.568.3_all.deb ...
Unpacking jenkins (2.568.3) ...
Setting up net-tools (2.10-2ubuntu1) ...
Setting up jenkins (2.568.3) ...
Created symlink '/etc/systemd/system/multi-user.target.wants/jenkins.service' + '/usr/lib/systemd/system/jenkins.service'.
Processing triggers for man-db (2.13.1-1build1) ...
Scanning processes...
Scanning linux images...

Running kernel seems to be up-to-date.

No services need to be restarted.

No containers need to be restarted.

No user sessions are running outdated binaries.

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-21-0:~$ sudo systemctl enable --now jenkins
Synchronizing state of jenkins.service with SysV service script with /usr/lib/systemd/systemd-sysv-install.
Executing: /usr/lib/systemd/systemd-sysv-install enable jenkins
ubuntu@ip-172-31-21-0:~$

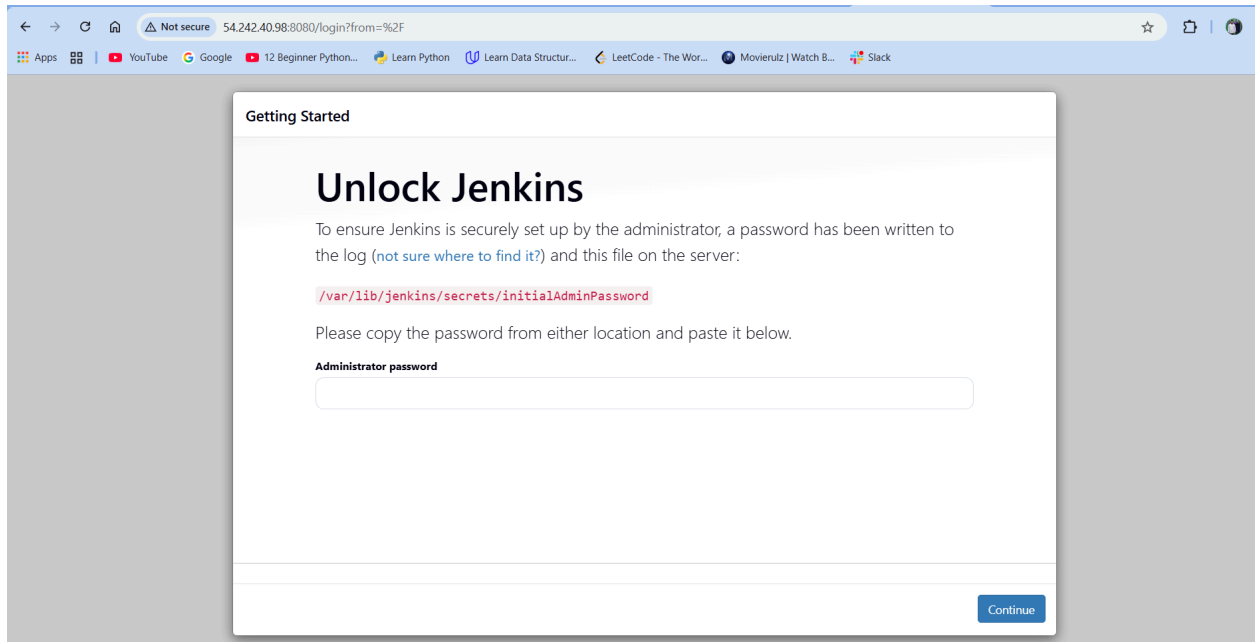
```

```

ubuntu@ip-172-31-21-0:~$ sudo systemctl status jenkins
● Jenkins.service - Jenkins Continuous Integration Server.
   Loaded: loaded (/usr/lib/systemd/system/jenkins.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-17 07:26:17 UTC; 1min 18s ago
   Invocation: 2f26394e35e64fc689d5ea3fb9d7d8c2
   Main PID: 4750 (java)
   Tasks: 37 (limit: 627)
   Memory: 316M (peak: 360.7M)
   CPU: 21.546s
   CGroup: /system.slice/jenkins.service
           └─4750 /usr/bin/java -Djava.awt.headless=true -jar /usr/share/java/jenkins.war --webroot=/var/cache/jenkins/war --httpPort=8080

Sep 17 07:26:12 ip-172-31-21-0 jenkins[4750]: [LF]>
Sep 17 07:26:12 ip-172-31-21-0 jenkins[4750]: [LF]> *****
Sep 17 07:26:12 ip-172-31-21-0 jenkins[4750]: [LF]> *****
Sep 17 07:26:12 ip-172-31-21-0 jenkins[4750]: [LF]> *****
Sep 17 07:26:17 ip-172-31-21-0 jenkins[4750]: 2026-09-17 07:26:17.413+0000 [id=34] INFO jenkins.InitReactorRunner$1#onAttained: Completed initialization
Sep 17 07:26:17 ip-172-31-21-0 jenkins[4750]: 2026-09-17 07:26:17.448+0000 [id=24] INFO hudson.lifecycle.Lifecycle#onReady: Jenkins is fully up and running
Sep 17 07:26:17 ip-172-31-21-0 systemd[1]: Started jenkins.service - Jenkins Continuous Integration Server.
Sep 17 07:26:17 ip-172-31-21-0 jenkins[4750]: 2026-09-17 07:26:17.695+0000 [id=50] INFO h.n.DownloadService$Downloadable#load: Obtained the updated data file for hudson.tasks.Maven.MavenInstaller
Sep 17 07:26:17 ip-172-31-21-0 jenkins[4750]: 2026-09-17 07:26:17.695+0000 [id=50] INFO hudson.util.Retrier#start: Performed the action check updates server successfully at the attempt #1
Sep 17 07:26:22 ip-172-31-21-0 jenkins[4750]: 2026-09-17 07:26:22.564+0000 [id=69] WARNING h.n.DiskSpaceMonitorDescriptor#markNodeOfflineonOnline: Making Built-In Node offline temporarily due to
lines 1-2/21 (END)

```



2. Setup Apache Tomcat on Ubuntu Virtual Machine

Setup Apache Tomcat on Ubuntu Virtual Machine

1. Go to `/tmp` directory

```
cd /tmp
```

2. Create Tomcat User

Create a dedicated user for running Tomcat:

```
sudo useradd -m -U -d /opt/tomcat -s /bin/false tomcat
```

Verify the user:

```
id tomcat
```

3. Download Apache Tomcat

Download the Tomcat package:

```
wget  
https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
```

Verify the downloaded file:

```
ls -lh apache-tomcat-11.0.26.tar.gz
```

4. Create Tomcat Installation Directory

```
sudo mkdir -p /opt/tomcat
```

5. Extract Tomcat

Extract Tomcat into `/opt/tomcat`:

```
sudo tar -xvzf apache-tomcat-11.0.26.tar.gz -C /opt/tomcat --strip-components=1
```

Verify:

```
sudo ls -lah /opt/tomcat
```

Important Tomcat directories:

```
bin  
conf  
lib  
logs  
temp  
webapps  
work
```

6. Change Tomcat Ownership

Give ownership of the Tomcat installation to the Tomcat user:

```
sudo chown -R tomcat:tomcat /opt/tomcat
```

7. Set Execute Permissions

Make the Tomcat shell scripts executable:

```
sudo chmod +x /opt/tomcat/bin/*.sh
```

Verify:

```
sudo ls -l /opt/tomcat/bin/
```

8. Install/Verify Java

Check Java:

```
java -version
```

In my setup:

```
openjdk version "21.0.12"
```

Find the Java installation path:

```
readlink -f $(which java)
```

Java path:

```
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
```

Therefore:

```
JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64
```

9. Create Tomcat Systemd Service

Create the service file:

```
sudo nano /etc/systemd/system/tomcat.service
```

Add:

```
[Unit]
```

```
Description=Apache Tomcat Web Application Container
```

```
After=network.target
```

[Service]

Type=forking

User=tomcat

Group=tomcat

Environment="JAVA_HOME=/usr/lib/jvm/java-21-openjdk-amd64"

Environment="CATALINA_HOME=/opt/tomcat"

Environment="CATALINA_BASE=/opt/tomcat"

ExecStart=/opt/tomcat/bin/startup.sh

ExecStop=/opt/tomcat/bin/shutdown.sh

[Install]

WantedBy=multi-user.target

Save and exit.

10. Reload Systemd

sudo systemctl daemon-reload

11. Start Tomcat

sudo systemctl start tomcat

12. Check Tomcat Status

sudo systemctl status tomcat

Expected:

Active: active (running)

13. Enable Tomcat at Boot

```
sudo systemctl enable tomcat
```

This ensures Tomcat automatically starts after a server reboot.

14. Check Tomcat Port

Tomcat normally uses port **8080**.

```
sudo ss -lntp | grep 8080
```

In my environment, port 8080 was already being used by Jenkins, so Tomcat was configured to use another available port, such as **8081**.

To check the configured Tomcat port:

```
sudo grep -n "Connector port" /opt/tomcat/conf/server.xml
```

If using 8081:

```
sudo ss -lntp | grep 8081
```

15. Test Tomcat

For port 8081:

```
curl http://localhost:8081
```

If Tomcat is working, it returns the Tomcat web page HTML.

16. Access Tomcat from Browser

If running on an AWS EC2 instance and Tomcat uses port 8081:

```
http://<EC2-PUBLIC-IP>:8081
```

Make sure the EC2 Security Group allows:

Type: Custom TCP

Port: 8081

Source: My IP

Final Verification Commands

java -version

sudo systemctl status tomcat

sudo systemctl is-enabled tomcat

sudo ss -lntp | grep 8081

curl http://localhost:8081

```
ubuntu@ip-172-31-21-0: /tmp
ubuntu@ip-172-31-21-0:~$ cd /tmp
ubuntu@ip-172-31-21-0: /tmp$ sudo useradd -m -U -d /opt/tomcat -s /bin/false tomcat
ubuntu@ip-172-31-21-0: /tmp$ id tomcat
uid=1001(tomcat) gid=1001(tomcat) groups=1001(tomcat)
ubuntu@ip-172-31-21-0: /tmp$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.24/bin/apache-tomcat-11.0.24.tar.gz
--2026-09-17 07:49:02-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.24/bin/apache-tomcat-11.0.24.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 404 Not Found
2026-09-17 07:49:03 [600K, 404: Not Found]

ubuntu@ip-172-31-21-0: /tmp$ ls -lh apache-tomcat-11.0.24.tar.gz
ls: cannot access 'apache-tomcat-11.0.24.tar.gz': No such file or directory
ubuntu@ip-172-31-21-0: /tmp$ wget https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
--2026-09-17 07:50:44-- https://dlcdn.apache.org/tomcat/tomcat-11/v11.0.26/bin/apache-tomcat-11.0.26.tar.gz
Resolving dlcdn.apache.org (dlcdn.apache.org)... 151.101.2.132, 2a04:4e42::644
Connecting to dlcdn.apache.org (dlcdn.apache.org)[151.101.2.132]:443... connected.
HTTP request sent, awaiting response... 200 OK
Length: 14482233 (14M) [application/x-gzip]
Saving to: 'apache-tomcat-11.0.26.tar.gz'

apache-tomcat-11.0.26.tar.gz 100%[=====] 13.81M --.-KB/s in 0.05s

2026-09-17 07:50:44 (261 MB/s) - 'apache-tomcat-11.0.26.tar.gz' saved [14482233/14482233]

ubuntu@ip-172-31-21-0: /tmp$ ls -lh apache-tomcat-11.0.26.tar.gz
-rw-rw-r-- 1 ubuntu ubuntu 14M Sep 9 22:01 apache-tomcat-11.0.26.tar.gz
ubuntu@ip-172-31-21-0: /tmp$ sudo mkdir -p /opt/tomcat
ubuntu@ip-172-31-21-0: /tmp$ sudo tar -xvzf apache-tomcat-11.0.26.tar.gz -C /opt/tomcat --strip-components=1
apache-tomcat-11.0.26/conf/
apache-tomcat-11.0.26/conf/catalina.properties
apache-tomcat-11.0.26/conf/context.xml
apache-tomcat-11.0.26/conf/jaspic-providers.xml
apache-tomcat-11.0.26/conf/jaspic-providers.xsd
apache-tomcat-11.0.26/conf/logging.properties
apache-tomcat-11.0.26/conf/server.xml
apache-tomcat-11.0.26/conf/tomcat-users.xml
apache-tomcat-11.0.26/conf/web.xml
apache-tomcat-11.0.26/bin/
apache-tomcat-11.0.26/lib/
apache-tomcat-11.0.26/logs/
apache-tomcat-11.0.26/temp/
apache-tomcat-11.0.26/webapps/
apache-tomcat-11.0.26/webapps/ROOT/
apache-tomcat-11.0.26/webapps/ROOT/WEB-INF/
apache-tomcat-11.0.26/webapps/docs/
apache-tomcat-11.0.26/webapps/docs/META-INF/
apache-tomcat-11.0.26/webapps/docs/WEB-INF/
apache-tomcat-11.0.26/webapps/docs/WEB-INF/jsp/
apache-tomcat-11.0.26/webapps/docs/annotationapi/
```

```

rwxr-x--- 1 tomcat tomcat 3.9K Sep  9 21:36 setclasspath.sh
-rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 shutdown.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 shutdown.sh
-rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 startup.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 startup.sh
-rw-r----- 1 tomcat tomcat 50K Sep  9 21:36 tomcat-juli.jar
-rw-r----- 1 tomcat tomcat 461K Sep  9 21:36 tomcat-native.tar.gz
-rw-r----- 1 tomcat tomcat 3.6K Sep  9 21:36 tool-wrapper.bat
rwxr-x--- 1 tomcat tomcat 4.5K Sep  9 21:36 tool-wrapper.sh
-rw-r----- 1 tomcat tomcat 2.0K Sep  9 21:36 version.bat
rwxr-x--- 1 tomcat tomcat 1.9K Sep  9 21:36 version.sh
ubuntu@ip-172-31-21-0:/tmp$ sudo chown -R tomcat:tomcat /opt/tomcat
ubuntu@ip-172-31-21-0:/tmp$ sudo chmod +x /opt/tomcat/bin/*.sh
hmod: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
ubuntu@ip-172-31-21-0:/tmp$ sudo ls -l /opt/tomcat/bin/*.sh
ls: cannot access '/opt/tomcat/bin/*.sh': No such file or directory
ubuntu@ip-172-31-21-0:/tmp$ java -version
openjdk version "21.0.12" 2026-07-21
OpenJDK Runtime Environment (build 21.0.12-81-26.04-Ubuntu)
OpenJDK 64-Bit Server VM (build 21.0.12-81-26.04-Ubuntu, mixed mode, sharing)
ubuntu@ip-172-31-21-0:/tmp$ readlink -f $(which java)
/usr/lib/jvm/java-21-openjdk-amd64/bin/java
ubuntu@ip-172-31-21-0:/tmp$ sudo nano /etc/systemd/system/tomcat.service
ubuntu@ip-172-31-21-0:/tmp$ sudo systemctl daemon-reload
ubuntu@ip-172-31-21-0:/tmp$ sudo systemctl start tomcat
ubuntu@ip-172-31-21-0:/tmp$ sudo systemctl status tomcat
tomcat.service - Apache Tomcat Web Application Container
   Loaded: loaded (/etc/systemd/system/tomcat.service; disabled; preset: enabled)
   Active: active (running) since Thu 2026-09-17 08:10:04 UTC; 9s ago
 Invocation: 9802e80bd348d60c2cf52b6cb9be
   Process: 5722 ExecStart=/opt/tomcat/bin/startup.sh (code=exited, status=0/SUCCESS)
  Main PID: 5732 (java)
    Tasks: 17 (limit: 627)
   Memory: 90.7M (peak: 90.7M)
      CPU: 4.834s
   CGroup: /system.slice/tomcat.service
           └─5732 /usr/lib/jvm/java-21-openjdk-amd64/bin/java -Djava.util.logging.config.file=/opt/tomcat/conf/logging.properties -Djava.util.logging.manager=org.apache.juli.ClassLoaderLogManager -Djdk.tls.
Sep 17 08:10:04 ip-172-31-21-0 systemd[1]: Starting tomcat.service - Apache Tomcat Web Application Container...
Sep 17 08:10:04 ip-172-31-21-0 startup.sh[5722]: Tomcat started.
Sep 17 08:10:04 ip-172-31-21-0 systemd[1]: Started tomcat.service - Apache Tomcat Web Application Container.
ubuntu@ip-172-31-21-0:/tmp$ sudo systemctl enable tomcat
Created symlink '/etc/systemd/system/multi-user.target.wants/tomcat.service' → '/etc/systemd/system/tomcat.service'.
ubuntu@ip-172-31-21-0:/tmp$ sudo ss -ltnp | grep 8080
LISTEN 0      50          *:8080          *:.*      users:((("java",pid=4750,fd=9))
ubuntu@ip-172-31-21-0:/tmp$ curl http://localhost:8080
html<head>meta http-equiv=refresh content=1;url=/login?from=%2F'><script id='redirect' data-redirect-url='/login?from=%2F' src='/static/47c9b500/scripts/redirect.js'></script></head><body style='background-color:white; color:white;'>
Authentication required
If you are authenticated as: anonymous
Groups that you are in:
anonymous
Permission you need to have (but didn't): hudson.model.Hudson.Administer
-->
/body></html>
ubuntu@ip-172-31-21-0:/tmp$

```

Successfully installed Apache Tomcat 11 on an Ubuntu Virtual Machine. Created a dedicated Tomcat user, downloaded and extracted Tomcat under `/opt/tomcat`, configured ownership and permissions, configured Java 21, created a systemd service, started and enabled Tomcat, and verified the Tomcat service and application port. Since Jenkins was already using port 8080, Tomcat was configured to use a separate port.

3. Setup Docker Desktop on your local windows.



Install Docker on AWS EC2 Ubuntu Instance

1. Launch EC2 Instance

- Region: **us-east-1 (N. Virginia)**
- Instance name: **EC2_Docker**
- AMI: **Ubuntu Server 22.04 LTS**
- Instance type: **t2.micro**
- Key pair: **my_server.pem**
- Security group: allow **SSH (22)** and any custom app ports (e.g., 8080, 5000).

2. Connect via SSH

```
bash
ssh -i "my_server.pem" ubuntu@<EC2-Public-IP>
```

3. Update Packages

```
bash
sudo apt update
sudo apt upgrade -y
```

4. Install Prerequisites

```
bash
sudo apt install ca-certificates curl gnupg lsb-release -y
```

5. Add Docker's Official GPG Key

```
bash
sudo mkdir -p /etc/apt/keyrings
curl -fsSL https://download.docker.com/linux/ubuntu/gpg | \
sudo gpg --dearmor -o /etc/apt/keyrings/docker.gpg
```

6. Set Up Docker Repository

```
bash
echo \
"deb [arch=$(dpkg --print-architecture) signed-by=/etc/apt/keyrings/docker.gpg] \
https://download.docker.com/linux/ubuntu \
$(lsb_release -cs) stable" | \
sudo tee /etc/apt/sources.list.d/docker.list > /dev/null
```

7. Install Docker Engine

```
bash
```

```
sudo apt update
```

```
sudo apt install docker-ce docker-ce-cli containerd.io -y
```

8. Start and Enable Docker

```
bash
```

```
sudo systemctl start docker
```

```
sudo systemctl enable docker
```

9. Verify Installation

```
bash
```

```
docker --version
```

```
sudo systemctl status docker
```

10. Run Test Container

```
bash
```

```
sudo docker run hello-world
```

👉 Confirms Docker is working.



Final Verification

Check running containers

```
bash
```

```
sudo docker ps
```

-

Check images

```
bash
```

```
sudo docker images
```

-

- **Run Ubuntu container**

```
bash
```

```
ubuntu@ip-172-31-21-0:~$ sudo apt update
sudo apt upgrade -y
Hit:1 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute InRelease
Get:2 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates InRelease [137 kB]
Hit:3 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-backports InRelease
Hit:4 https://pkg.jenkins.io/debian-stable binary/ InRelease
Hit:5 https://pkg.jenkins.io/debian-stable binary/ Release
Get:6 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/main amd64 Packages [614 kB]
Get:7 http://us-east-1.ec2.archive.ubuntu.com/ubuntu resolute-updates/universe amd64 Packages [295 kB]
Hit:9 http://security.ubuntu.com/ubuntu resolute-security InRelease
Fetched 1046 kB in 0s (3197 kB/s)
173 packages can be upgraded. Run 'apt list --upgradable' to see them.
The following package was automatically installed and is no longer required:
  pollinate
Use 'sudo apt autoremove' to remove it.

Upgrading:
  apparmor          dracut          grub2-common      libfwupd3          libpython3.14      mdadm              python3-idna        ubuntu-kernel-accessories
  apparmor          dracut-core      grub              libgccrypt2        libpython3.14-minimal  ntdm-news-config  python3-problem-report  ubuntu-minimal
  apparmor-core-dump-handler  dracut-install  inetutils-telnet  libgssapi-krb5-2   libpython3.14-stdlib  mount              python3-pyasn1       ubuntu-pro-client
  base-files        dracut-network  iproute2          libk5crypto3       libsgutil2-1.48     nano               python3.14          ubuntu-pro-client-110n
  bind9-dnsutils    eject           keyboard-configuration  libkrb5-3          libsmartcols1       nftables-3p       python3.14-gdbm     ubuntu-release-upgrader-core
  bind9-host        fdisk           kbd-locale        libkrb5support0    libsqlite3-0        openssl-client     python3.14-minimal  ubuntu-server
  bind9-libs        fwupd           libapparmor1      liblastlog2-2      libssh2-1t64        openssh-server     rsync               ubuntu-standard
  bpftool           gawk           libattr1          libmount1          libssl3t64          openssl-sftp-server  rsynclog            udev
  bsdextrautils     gir1.2-packagekitglib-1.0  libaudit1-common  libnftables1       libsystemd-shared   openssl-provider-legacy  sg3-utils           udisks2
  bsdutils          gnucoreutils   libaudit1         libnss-systemd     libsystemd0          sg3-utils-udev      update-notifier-common  util-linux
  ca-certificates  gnupg          libblkid1         libnss3            libsystemd-shared   sos                vim                  util-linux-extra
  cloud-init        gnupg-l10n     libbz2-1.0        libnss-sqlite3     libsystemd-sysv     sudo-rs            vim-tiny             xfs
  cloud-init-base   gnupg-utils    libc-bin          libnss-systemd     libsystemd-timesyncd  sudo-rs            xfs                   xz
  console-setup     gpg            libc-bin          libpam-modules     libsystemd-timesyncd  sudo-rs            xz                   xz
  console-setup-l10n  gpg-agent      libc-bin          libpam-modules-bin  libsystemd-timesyncd  sudo-rs            xz                   xz
  cpio              gpg-wks-client  libcurl4t64-gnutls  libpam-runtime     libsystemd-timesyncd  sudo-rs            xz                   xz
  curl              gpgconf        libcurl4t64       libpam-systemd     libsystemd-timesyncd  sudo-rs            xz                   xz
  diffutils         gpgsm          libevent-core-2.1-7t64  libpam-systemd     libsystemd-timesyncd  sudo-rs            xz                   xz
  dirsync           gpgv           libfdisk1         libperl5.40        libsystemd-timesyncd  sudo-rs            xz                   xz
  distro-info-data  grub-pc        libflac12t64      libperl5.40        libsystemd-timesyncd  sudo-rs            xz                   xz
  dmidecode         grub-pc-bin    libfreebyte6       libpolkit-agent-1-0  libsystemd-timesyncd  sudo-rs            xz                   xz
  dmidecode         grub-pc-bin    libfreebyte6       libpolkit-gobject-1-0  libsystemd-timesyncd  sudo-rs            xz                   xz

Installing dependencies:
  linux-aws-headers-7.0.0-1012  linux-headers-7.0.0-1012-aws  linux-main-modules-zfs-7.0.0-1012-aws  linux-tools-7.0.0-1012-aws
  linux-aws-tools-7.0.0-1012  linux-image-7.0.0-1012-aws  linux-modules-7.0.0-1012-aws

Suggested packages:
  linux-aws-doc-7.0.0 | linux-aws-source-7.0.0 | linux-aws-tools

Not upgrading yet due to phasing:
  libnetplan1 netplan-generator netplan.io python3-netplan python3-software-properties software-properties-common

Summary:
  Upgrading: 167, Installing: 7, Removing: 0, Not Upgrading: 6
  127 standard LTS security updates
  Download size: 344 MB
  Space needed: 298 MB / 3889 MB available
```

```
ubuntu@ip-172-31-21-0:~$ sudo systemctl status docker
● docker.service - Docker Application Container Engine
   Loaded: loaded (/usr/lib/systemd/system/docker.service; enabled; preset: enabled)
   Active: active (running) since Thu 2026-09-17 09:28:43 UTC; 1min 6s ago
   Invocation: a05f6de0faa74c43b6af22cdd763df9
   Triggers: ● docker.socket
   Docs: https://docs.docker.com
   Main PID: 36788 (dockerd)
   Tasks: 0
   Memory: 34.1M (peak: 107.5M)
   CPU: 747ms
   CGroup: /system.slice/docker.service
           └─36788 /usr/bin/dockerd -H fd:// --containerd=/run/containerd/containerd.sock

Sep 17 09:28:42 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:42.910087029Z" level=info msg="Restoring containers: start."
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.002802832Z" level=info msg="Deleting nftables IP4 rules" error="running nft: /dev/stdin:1:17-30: Error: Could not process rule: No such file or directory"
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.010530898Z" level=info msg="Deleting nftables IP6 rules" error="running nft: /dev/stdin:1:18-31: Error: Could not process rule: No such file or directory"
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.378478680Z" level=info msg="Loading containers: done."
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.406315147Z" level=info msg="Docker daemon" commit=64cd50 containerd-snapshotter=true storage-driver=overlays version=29.8.1
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.407117402Z" level=info msg="Initializing buildkit"
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.762497094Z" level=info msg="Completed buildkit initialization"
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.774022949Z" level=info msg="Daemon" commit=64cd50 containerd-snapshotter=true storage-driver=overlays version=29.8.1
Sep 17 09:28:43 ip-172-31-21-0 dockerd[36788]: time="2026-09-17T09:28:43.774254990Z" level=info msg="API listen on /run/docker.sock"
Sep 17 09:28:43 ip-172-31-21-0 systemd[1]: Started docker.service - Docker Application Container Engine.

ubuntu@ip-172-31-21-0:~$ sudo docker run hello-world
Unable to find image 'hello-world:latest' locally
latest: Pulling from library/hello-world
4ff5586f7dd0: Pull complete
Digest: sha256:5e239053324d887c48ad5e5c56d294eab81588df9605b07d1afe895f9cc8f8
Status: Downloaded newer image for hello-world:latest

Hello from Docker!
This message shows that your installation appears to be working correctly.

To generate this message, Docker took the following steps:
 1. The Docker client contacted the Docker daemon.
 2. The Docker daemon pulled the "hello-world" image from the Docker Hub.
    (amd64)
 3. The Docker daemon created a new container from that image which runs the
    executable that produces the output you are currently reading.
 4. The Docker daemon streamed that output to the Docker client, which sent it
    to your terminal.

To try something more ambitious, you can run an Ubuntu container with:
$ docker run -it ubuntu bash

Share images, automate workflows, and more with a free Docker ID:
https://hub.docker.com/

For more examples and ideas, visit:
https://docs.docker.com/get-started/

ubuntu@ip-172-31-21-0:~$
```

